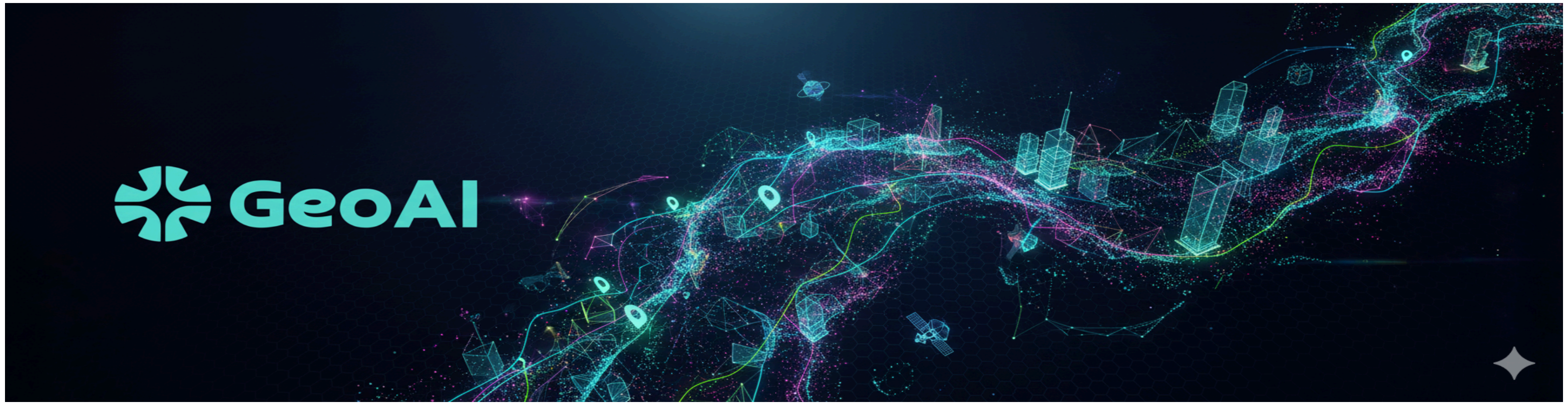
GeoAI

Geospatial Graph Learning

Course Intro — how this whole course works

~15 min · delivered once, before Unit 1



The Arena
Upskilling Your Knowledge

Your instructor — Ben Galon

Background & education

- **Technion Ph.D., Geo-Information Engineering** — thesis on bridging social networks and spatial data. *Graphs over geography, literally.*
- **Technion B.Sc.**, Mapping & Geo-Information — *Cum Laude*.
- 15+ years turning spatial & graph data into shipped products.

Work

- **Head of Core, Line5** — current.
- **VP Data, Lumen** — self-service AI, metrics, the product's data pipeline.
- **VP Data & R&D, Namogoo** — 35-person org, **billions of events/day**.
- **Head of Data Science, Moovit → Intel** — mobility analytics, trip-planning, real-time transit. *This course's world was my day job.*

Still active in research — recent work on trajectory matching & spatial big data (ACM SIGSPATIAL · WWW).

First: a 2-minute pulse check

Scan and answer — **anonymous**, no login, ~2 minutes. We'll watch the results fill in **live**, then read them together.

- 12 quick questions: how you **use AI, write code, and check its work** — in 2021, today, and (later) after this course.
- The real one: can you **hand an AI a task, supervise it, and check the result** — or do you still work like a programmer from a few years ago?

Scan to answer

Live counter on the screen →

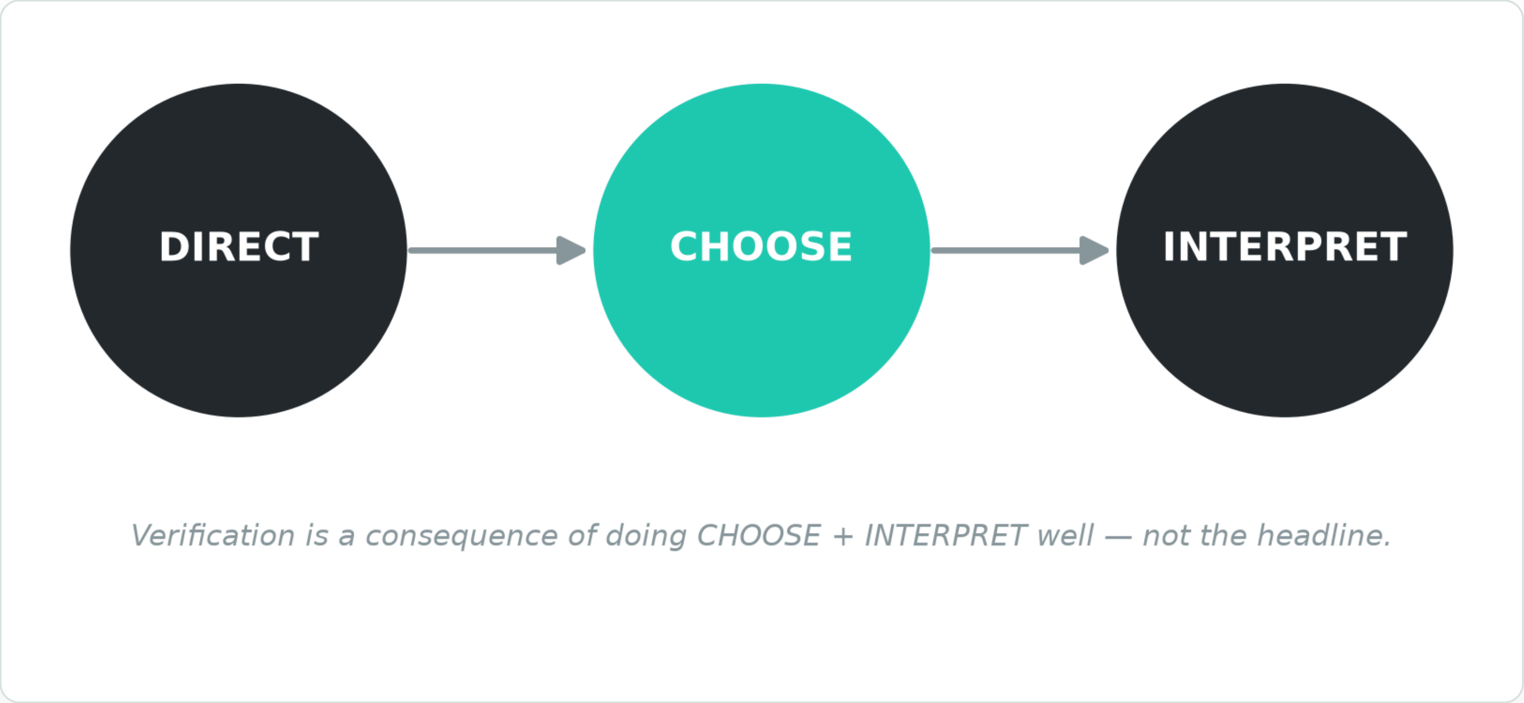
> No wrong answers — this is a **snapshot of the room**, not a test.

Why this course exists

- **Graph + Geo + AI** is a genuine, under-exploited opportunity: cities, road networks, sensor grids, and trajectories are all naturally **graphs**.
- You will not out-type the AI. Your edge is **knowing what to ask for** and **knowing whether the answer is right**.
- This course makes you the **analyst who directs**, not the one who memorizes API calls.

> **How you'd ask an AI:** "*I have a city road network — what graph representations and metrics let me reason about it? Walk me through the modeling choices.*"

The North Star: DIRECT → CHOOSE → INTERPRET

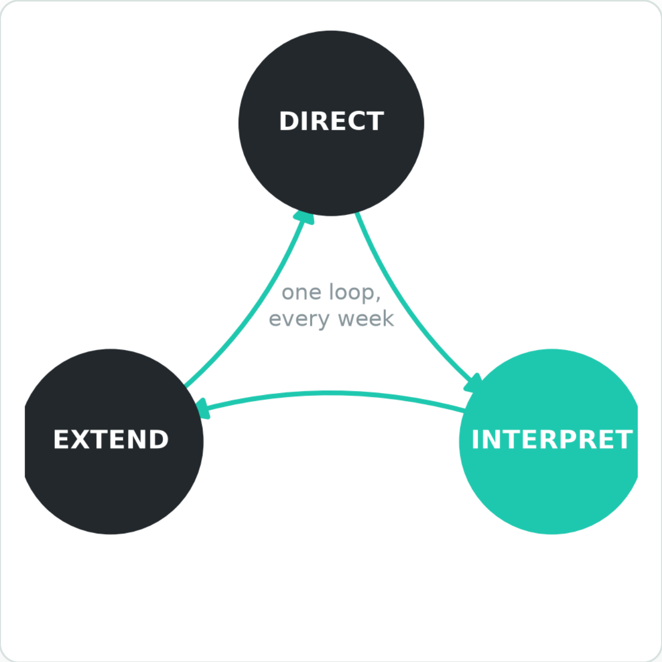


1. **DIRECT** an AI using **domain-precise vocabulary**.
2. **CHOOSE** the right analysis for a **real question**.
3. **INTERPRET** results in **domain terms** — spot surprises worth chasing.

The loop you rehearse every week

The third hour of every unit is **supervised practice** — one loop, run again
and again:

- **DIRECT** — phrase the request in the unit's vocabulary, with precise inputs.
- **INTERPRET** — restate the result in your own domain words; flag surprises.
- **EXTEND** — ask the follow-up the result just earned.



> **How you'd ask an AI:** "*Restate what this result implies in domain terms, then propose the single most useful follow-up.*"

Two skills we won't teach — but good practice needs

The units give you the **domain vocabulary**. Two *meta-skills* sit underneath every good session — and they're on **you** to build:

Prompt engineering

Precise DIRECTing *is* prompting: give context, name constraints, ask for the **reasoning**, show an example of the shape you want. A vague ask gets a vague answer — in any unit's vocabulary.

Working in loops

The first answer is rarely *the* answer. **Run → read → refine the ask → run again.** The **direct → interpret → extend** cycle is exactly this discipline, made routine.

> Neither lives in a unit's theory block — but they're the difference between a frustrating hour and a productive one. Build them as you go.

The single rubric — used every supervised hour

DIRECTING — Did I use this unit's vocabulary? Did I specify inputs precisely (which city, which metric, which window)?

INTERPRETING — Can I explain the result in my own domain words? Did I notice anything that didn't match my expectation?

EXTENDING — Did I ask a follow-up grounded in what the result revealed?

> Five checkboxes, three verbs. You should recite it from memory by Unit 3.
> *(This is `rubric.md`, the same in all five units — only the vocabulary changes.)*

The decision log — your reasoning, on the record

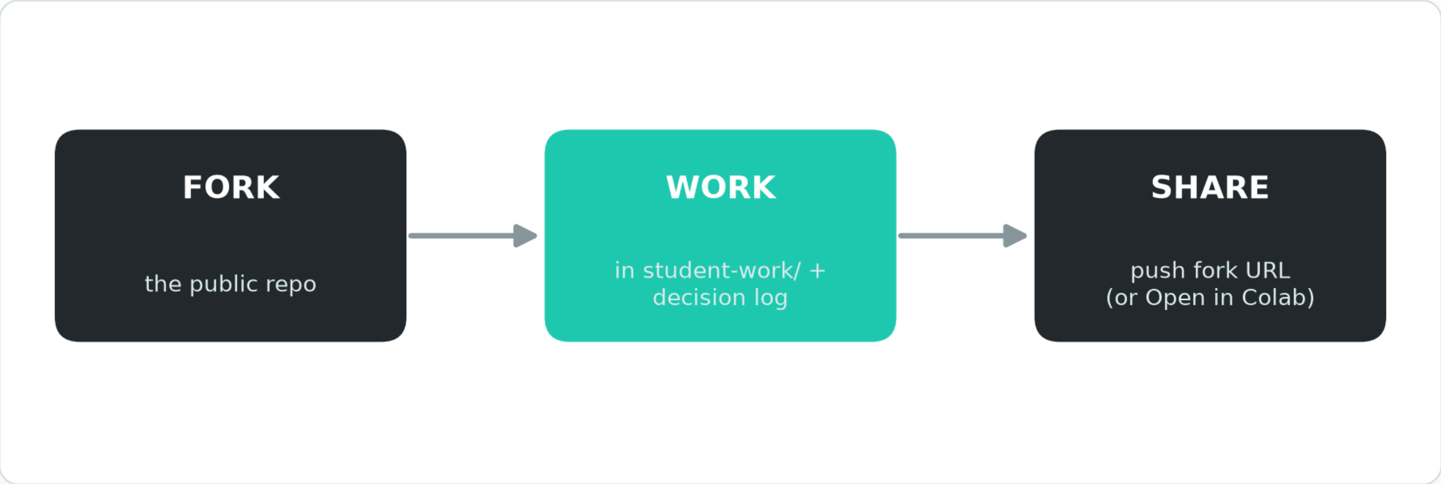
For every direct → interpret → extend cycle, you write **one entry**:

- What you **asked** the agent (and why, in this unit's vocabulary).
- What it **returned**, and how you **read** it.
- The **follow-up** it provoked.

Copy `decision-log-template.md` into your working folder. The log — not a polished notebook — is the deliverable for the in-class hour.

> **How you'd ask an AI:** "*Summarize the last three things we did into a decision-log entry: my request, your result, my interpretation, next question.*"

The workflow: fork, work, share



- **Fork** the public repo; install Claude Code; `uv sync --extra unit-1`; run the smoke-test cell.
- Your work lives in `student-work/` — conflict-free in your fork.
- A **reference solution ships day 1** — *one* path, not *the* answer. Read it after you've tried.
- **Async help:** push your fork + decision log; "Open in Colab" is the fallback.

The five units — your capability arc

#	Unit	After it, you can…	Status
1	**Graph Substrate**	Turn a road map into a graph; read topology metrics as signal.	✅ ready
2	**Trajectory Mining**	Put noisy GPS *on and off* the graph; separate signal from noise.	✅ ready
3	**Statistical Baselines**	Forecast a sensor; find the *breakeven horizon* + the spatial gap.	✅ ready
4	**Dynamic Navigation**	Route under time-varying cost; quantify multimodal "Cost of Anarchy."	🔬 preview
5	**Spatio-Temporal GNNs**	Build an ST-GNN that *earns* its complexity; interrogate its attention.	🔬 preview

One thread runs through it: a city **bus archive** — sparse GPS that U2 turns into routes, U3/U4 into speed forecasts and routing, U5 into a learned predictor.

How the course reaches you

- The full course is **5 units**. Units **1–3 are final**; **4–5 are preview** — the demo runs, but expect changes before we teach them.
- Lessons publish as **tagged releases** (`v-unit1`, `v-unit2`, …). **Sync your fork** to pull each new unit as it lands.
- `SYLLABUS.md` in the repo is your map: per-unit question, capability, status, and schedule. Read it first.

Every unit, same shape:

- **Theory** 45m — the deck
- **Demo** 45m — the notebook
- **Practice** 60m — *your* direct → interpret → extend loop

Each practice also hides one **"wrong-class" question** — one this unit's tools should *not* answer. Spotting it is half the skill.

Now: Unit 1 — The Graph Substrate

Looking only at the road map, which streets are the arteries?

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Rights & disclaimer

Built with AI assistance. These slides, notebooks, and notes were drafted with AI assistance (Claude Code) and **reviewed by a human instructor** — but they **may still contain mistakes**. **Verify before you rely on anything here.**

> Practice tasks are open-ended: a reference solution shows *a* strong path, not *the* answer.